

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Ans.

Devices using SMPS:

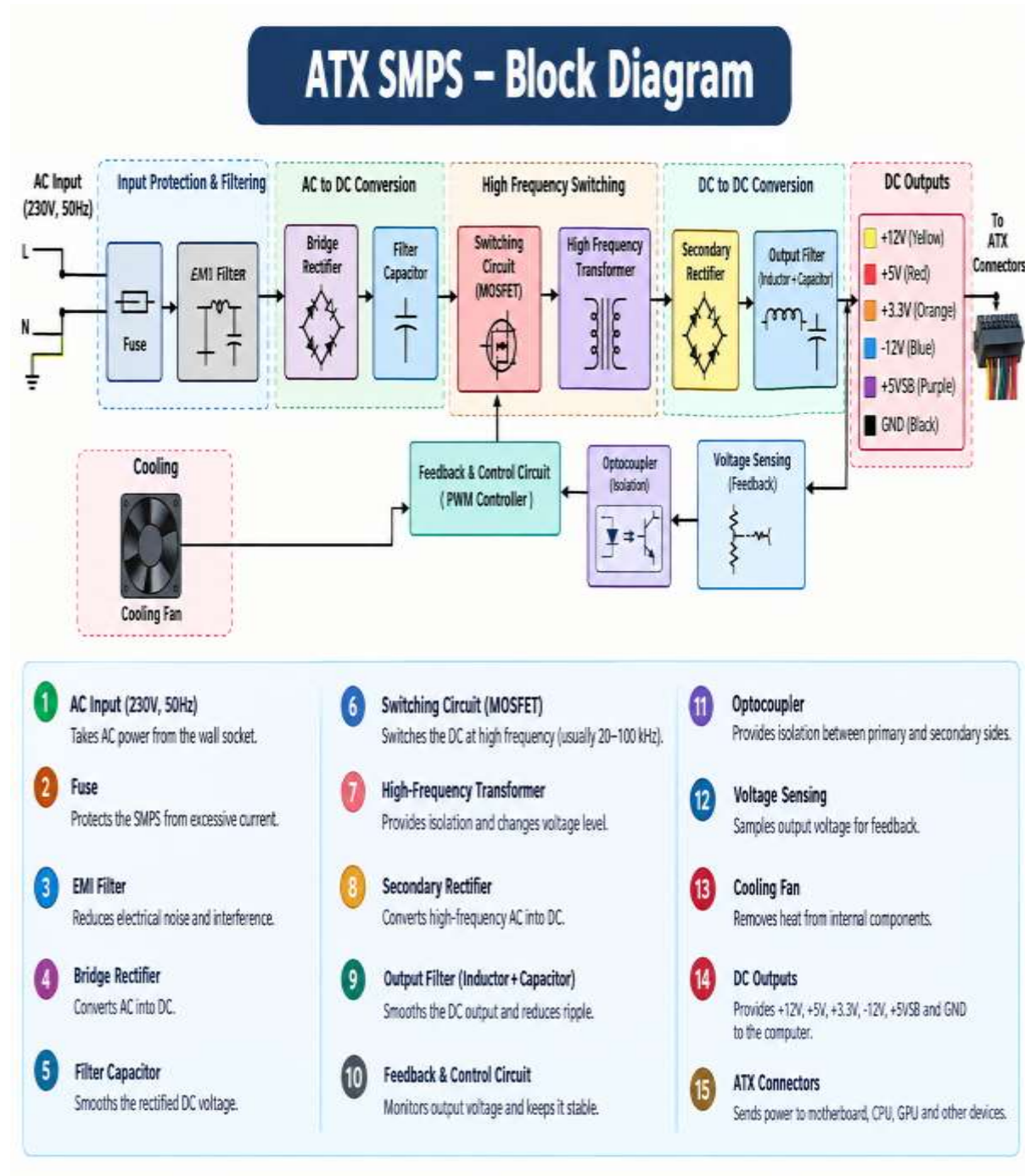
1. **Desktop Computer:** SMPS converts AC power into the required DC power for computer components.
2. **Wi-Fi Router:** SMPS provides the required low-voltage DC power to the router.
3. **LED Monitor:** SMPS converts AC power into suitable DC power for the monitor.

Devices using UPS:

1. **Desktop Computer:** UPS provides backup power during a power cut and prevents sudden shutdown.
2. **CCTV System:** UPS keeps the CCTV cameras and recording system working during a power failure.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Ans.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

Ans.



CONNECTOR NAME	DEVICES IT POWERS	SAFETY TIP
ATX 24-pin Motherboard Connector	This connector is used to supply power to the motherboard . It is the main power connector of a desktop computer.	Always switch off and unplug the PC before connecting or removing it. Make sure the clip is properly locked.
SATA Power Connector	It provides power to HDD, SSD and DVD/CD drives . It is commonly used in modern desktop computers.	Insert it in the correct direction and do not force the connector. Remove it gently when needed.
Molex 4-Pin Connector	It is mostly used with older hard drives, DVD drives and cooling fans . It was commonly used in older desktop PCs.	Turn off and unplug the PC before removing it. Hold the connector while pulling instead of pulling the wires.

Peripheral Troubleshooting Theory

PCIe 6/8-Pin Connector	Provides additional power to a PCIe graphics card (GPU).	Use the correct PCIe power cable and make sure it is fully connected before starting the computer.
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4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Ans.

1. **Switch off the power** – First, turn off the PC and unplug the power cable from the wall socket.
2. **Install the SMPS** – Place the SMPS inside the computer cabinet and fix it properly with screws.
3. **Connect the motherboard** – Take the 24-pin ATX power cable from the SMPS and connect it to the 24-pin power socket on the motherboard.
4. **Connect the CPU power** – Connect the 4-pin or 8-pin CPU/EPS cable from the SMPS to the CPU power connector near the processor.
5. **Connect the graphics card** – If the gaming PC has a dedicated GPU, connect the PCIe 6-pin, 8-pin, or 6+2-pin cable from the SMPS to the graphics card.
6. **Connect the storage drive** – For a SATA HDD or SATA SSD, connect the SATA power cable from the SMPS to the power connector on the drive.
7. **Connect other devices** – If any older device or fan needs power, use the Molex connector if provided by the SMPS.
8. **Check all connections** – Make sure all connectors are properly fitted and their clips are locked.
9. **Final check** – Check that no cable is loose and that the cables are away from the cooling fans.
10. **Start the PC** – Connect the main AC power cable to the SMPS, switch it on, and then start the computer.

Main Cables Used

- **24-pin ATX cable → Motherboard**
- **4/8-pin CPU/EPS cable → CPU power**
- **PCIe 6/8-pin or 6+2-pin cable → Graphics Card**
- **SATA power cable → HDD/SSD**
- **Molex cable → Older devices/Fans**